

Chapter 1: Number Systems & Fundamentals

Real & Complex Numbers, Interval Notation, Order of Operations, Absolute Value & Inequalities

1.1 Classification of Real Numbers

The set of real numbers $\mathbb{R}$ encompasses all rational and irrational numbers. Understanding this hierarchy is essential for analysis, algebra, and calculus foundations.

Set	Symbol	Definition & Examples
Natural Numbers	$\mathbb{N}$	Counting numbers: {1, 2, 3, 4, ...}
Integers	$\mathbb{Z}$	Whole numbers and negatives: {..., -2, -1, 0, 1, 2, ...}
Rational Numbers	$\mathbb{Q}$	Ratios of integers a/b ($b \neq 0$): e.g., $3/4$, -5, 0.333...
Irrational Numbers	$\mathbb{I}$ or $\mathbb{R} \setminus \mathbb{Q}$	Non-repeating, non-terminating decimals: e.g., $\sqrt{2}$, π , e
Complex Numbers	$\mathbb{C}$	Numbers in the form $a + bi$, where $i = \sqrt{-1}$

1.2 Absolute Value & Properties

Definition of Absolute Value

The absolute value of a real number x represents its distance from zero on the real line:

$$\begin{aligned} |x| &= x \text{ if } x \geq 0 \\ |x| &= -x \text{ if } x < 0 \end{aligned}$$

Core Algebraic Properties

- $|a \cdot b| = |a| \cdot |b|$
- $|a / b| = |a| / |b|$ ($b \neq 0$)
- Triangle Inequality:** $|a + b| \leq |a| + |b|$
- $|x| \leq a \Leftrightarrow -a \leq x \leq a$

Worked Example 1.1: Solving Absolute Value Inequalities

Problem: Solve for x : $|2x - 5| \leq 9$.

Solution: Rewrite as a compound inequality:

$-9 \leq 2x - 5 \leq 9 \Rightarrow$ Add 5 to all parts: $-4 \leq 2x \leq 14 \Rightarrow$ Divide by 2: $-2 \leq x \leq 7$.

Interval Notation: $[-2, 7]$.

1.3 Interval Notation & Inequalities

Intervals express subsets of real numbers. Square brackets $[]$ include endpoints (closed), while parentheses $()$ exclude endpoints (open).

Types of Intervals

- Open: $(a, b) \Rightarrow \{x \in \mathbb{R} \mid a < x < b\}$
- Closed: $[a, b] \Rightarrow \{x \in \mathbb{R} \mid a \leq x \leq b\}$
- Half-Open: $[a, b) \Rightarrow \{x \in \mathbb{R} \mid a \leq x < b\}$
- Infinite: $(-\infty, a] \Rightarrow \{x \in \mathbb{R} \mid x \leq a\}$

Inequality Rules

1. Multiplying/dividing by a negative number **reverses** the inequality sign.
2. If $a > b$ and $b > c$, then $a > c$ (Transitivity).
3. Reciprocal rule: If $a > b > 0$, then $1/a < 1/b$.

1.4 Introduction to Complex Numbers

Complex numbers extend the real number system by introducing the imaginary unit $i = \sqrt{-1}$, satisfying $i^2 = -1$.

Standard Form: $z = a + bi$ ($a = \text{Re}(z)$, $b = \text{Im}(z)$)

Complex Conjugate: $\bar{z} = a - bi$

Modulus (Magnitude): $|z| = \sqrt{a^2 + b^2}$

Worked Example 1.2: Complex Division

Problem: Express $(3 + 2i) / (1 - i)$ in standard form $a + bi$.

Solution: Multiply numerator and denominator by the conjugate of the denominator $(1 + i)$:

$$\begin{aligned} (3 + 2i)(1 + i) / [(1 - i)(1 + i)] &= (3 + 3i + 2i + 2i^2) / (1^2 - i^2) \\ &= (3 + 5i - 2) / (1 + 1) = (1 + 5i) / 2 = \mathbf{1/2 + (5/2)i}. \end{aligned}$$

Chapter 1 Review & Practice Problems

1. Classify each of the following numbers into all applicable sets ($\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{I}$, $\mathbb{C}$): **a)** $-7/3$, **b)** $\sqrt{5}$, **c)** 0 , **d)** $4 + 3i$.
2. Solve the absolute value inequality and express the solution in interval notation: $|3x + 2| > 8$.
3. Simplify the complex expression: $(2 - 4i)(3 + 5i) - (1 + i)^2$.
4. Find the modulus and complex conjugate of $z = 5 - 12i$.

